

SEX DETERMINATION

The mouse *Sry* locus harbors a cryptic exon that is essential for male sex determination

Shingo Miyawaki^{1,2}, Shunsuke Kuroki^{1,2}, Ryo Maeda^{1,2}, Naoki Okashita^{1,2}, Peter Koopman³, Makoto Tachibana^{1,2*}

The mammalian sex-determining gene *Sry* induces male development. Since its discovery 30 years ago, *Sry* has been believed to be a single-exon gene. Here, we identified a cryptic second exon of mouse *Sry* and a corresponding two-exon type *Sry* (*Sry-T*) transcript. XY mice lacking *Sry-T* were sex-reversed, and ectopic expression of *Sry-T* in XX mice induced male development. *Sry-T* messenger RNA is expressed similarly to that of canonical single-exon type *Sry* (*Sry-S*), but SRY-T protein is expressed predominantly because of the absence of a degron in the C terminus of SRY-S. *Sry* exon2 appears to have evolved recently in mice through acquisition of a retrotransposon-derived coding sequence to replace the degron. Our findings suggest that in nature, SRY-T, not SRY-S, is the bona fide testis-determining factor.

Sexual differentiation is essential for the survival and evolution of a species. Expression of the Y chromosomal gene *Sry* is required for male development in mammals (1). Since its discovery in 1990 (2, 3), *Sry* has been considered a single-exon gene, containing only one open reading frame (ORF) and encoding one protein. The mouse *Sry* locus has a complex genomic structure, with 2.8 kb of specific sequence containing an ORF, flanked by ~50 kb of palindromic sequences (4). Using a comprehensive transcriptomics approach, we

identified a sequence within the flanking palindromic sequence that is essential for male development.

We previously developed a method to enrich somatic cell populations from mouse embryonic gonads (5, 6) (fig. S1A). Using this method, NR5A1-expressing mouse pre-Sertoli cells were purified for RNA-sequencing (RNA-seq) analysis. We identified a transcribed sequence located within the palindromic sequences 3' to the known *Sry* ORF (Fig. 1A and fig. S1B). This sequence was transcribed similarly to the known *Sry* transcript in terms of

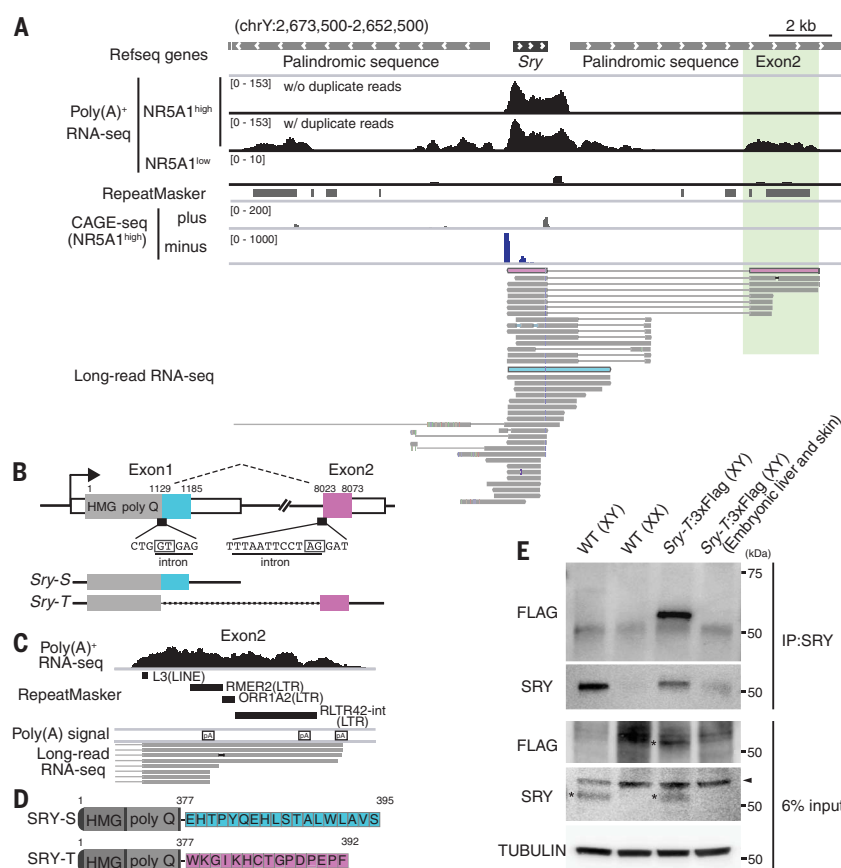
temporal and spatial specificity (fig. S1, C and D) but lacked a cap site, as evidenced by cap analysis gene expression sequencing (CAGE-seq) (Fig. 1A and fig. S1B). Long-read RNA-seq identified this sequence as the 3' portion of a new *Sry* transcript that starts at the known *Sry* transcription start site and is spliced once at typical 5' GT/3' AG splicing site sequences with an internal polypyrimidine tract (Fig. 1B and fig. S1, E and F). Thus, this sequence represents a second exon of *Sry*.

Sequencing data indicated the presence of two mRNA transcripts, the known single-exon type (*Sry-S*) and a two-exon type (*Sry-T*) (Fig. 1B and fig. S2). Most (~70%) of *Sry* exon2 is occupied by one long interspersed nuclear element (LINE) L3 and three long terminal repeats (LTRs) (Fig. 1C and fig. S3). *Sry-T* expression levels were approximately half those of *Sry-S* in pre-Sertoli cells (fig. S1G) and, as evidenced by single-cell transcriptome analysis, individual pre-Sertoli cells expressed both transcripts (7) (fig. S1H).

The predicted SRY-T protein shares amino acids 1 to 377 with SRY-S, including the high-mobility group DNA-binding domain and polyglutamine (poly-Q) sequences that are important for transcriptional activation (8, 9). Beyond this, the C-terminal 15 amino acids of SRY-T are encoded by the second exon, whereas absence of

Fig. 1. A cryptic exon of mouse *Sry* generates a new transcript and protein.

(A) *Sry* exon2 is shown shaded green. The RefSeq *Sry* (NM_011564.1) encoded in the (–) strand is presented at the top. Poly(A)⁺ RNA-seq data were mapped with or without excluding potential duplicate reads. RepeatMasker shows transposon-derived sequences. CAGE-seq identified transcription start sites. Individual long-read RNA-seq reads (>1000 bp) were mapped. Thick and thin lines represent exonic and intronic sequences, respectively. Representative long-read RNA transcripts for the new two-exon *Sry-T* and canonical single-exon *Sry-S* are shown in red and blue, respectively. (B) Exon-intron structure of the *Sry* locus. Shaded and open boxes represent CDS and untranslated regions, respectively. Adenine of the *Sry* translation start codon is position 1. Sequences of *Sry* splice junctions are represented below the bold lines. 5' GT/3' AG splicing motifs are boxed. CDS specific for *Sry-S* (blue) and *Sry-T* (red) are shown. (C) Retrotransposon-derived sequences in *Sry* exon2. An L3-derived sequence accounts for exon2 CDS, whereas LTR-type retrotransposon-derived sequences provide poly(A) signals of *Sry-T*. (D) Predicted amino acid sequences of SRY isoforms. (E) Anti-SRY and anti-FLAG immunoblots using anti-SRY immune complexes prepared from E11.5 *Sry-T*:3xFlag gonads showing signals with the predicted size of SRY-T-3xFLAG (~55 kDa). Asterisks and arrowhead represent putative SRY signals and nonspecific signals, respectively.



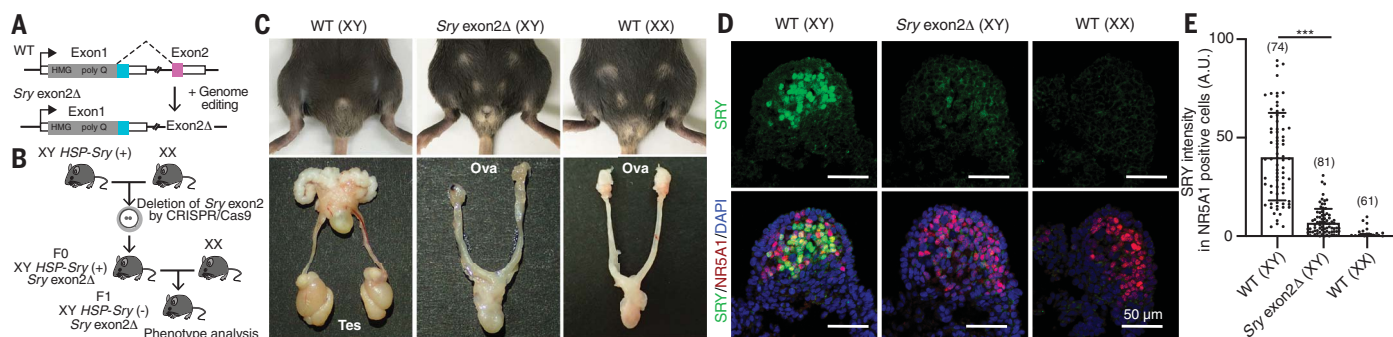
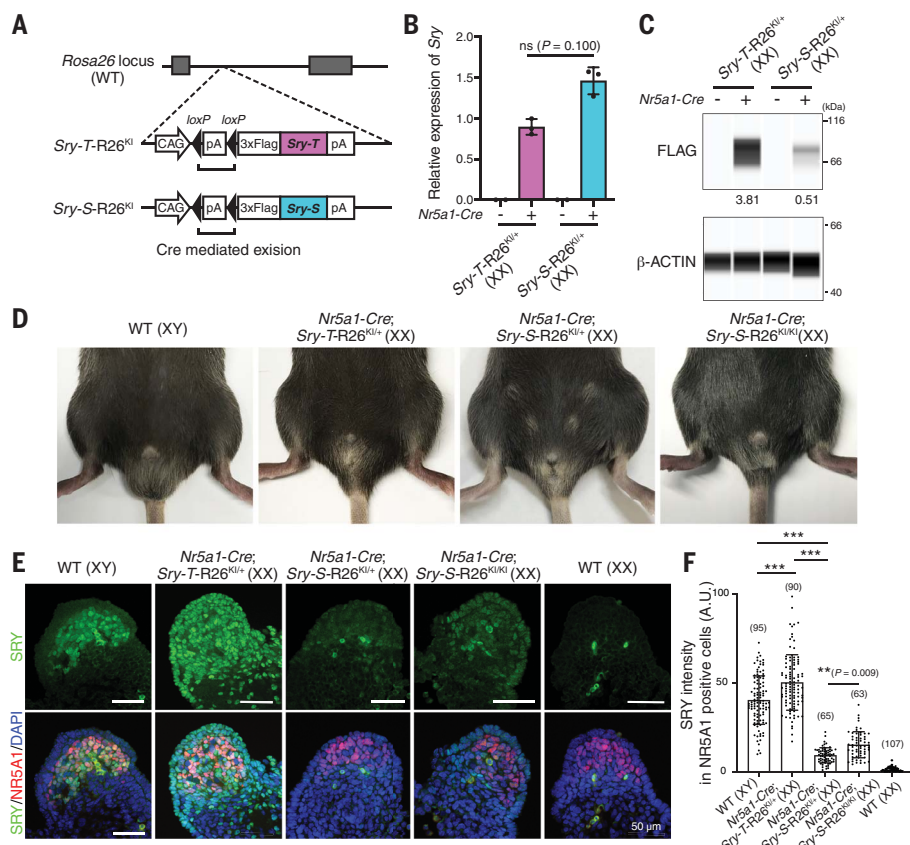


Fig. 2. *Sry-T* is necessary for male development in mice. (A) Deletion of *Sry* exon2 by CRISPR/Cas9. (B) Establishment of *Sry-T*-deficient mice. Procedures are detailed in fig. S5, A and B. To circumvent potential male infertility of *Sry-T*-deficient male mice, we used XY zygotes carrying the *HSP-Sry* transgene (17) for genome editing. The resultant F₁ generation without the *HSP-Sry* transgene was used for phenotypic analysis. (C) *Sry-T*-deficient mice exhibited male-to-female sex reversal ($n > 40$). External genitalia, nipples (top panels), and internal genitalia (bottom

panels) of adult mice are shown. Tes, testis; Ova, ovary. (D) Coimmunostaining profiles of SRY and the gonadal somatic marker NR5A1 in E11.5 gonads of the indicated genotypes. Data are representative of biological triplicate experiments. (E) Quantitation of SRY in NR5A1⁺ cells in E11.5 gonads of the indicated genotypes. A.U., arbitrary units. Data are shown as means $\pm$ SD. Numbers of analyzed cells are shown in parentheses above each bar. *** $P < 0.001$ between the indicated groups, one-way analysis of variance (ANOVA) with Tukey's post hoc test.

Fig. 3. SRY-T is the predominant SRY isoform and is sufficient to sex-reverse XX mice.

(A) Establishment of mouse lines carrying *Sry* knocked-in *Rosa26* alleles named *Sry-T-R26^{Kl}* and *Sry-S-R26^{Kl}*. Cre/loxP-mediated genome recombination induces *Sry* expression by removing the triple-poly(A) stop cassette between the promoter and the *Sry* CDS. (B) Relative mRNA expression of *Sry* in E11.5 gonads of the indicated genotypes. *Sry* expression was normalized to *Gapdh*. Data are shown as means $\pm$ SD. ns, not significant, Mann-Whitney *U* test. (C) Capillary-based immunodetection of E11.5 gonadal cell lysates of the indicated genotypes. Expression levels of FLAG relative to β -ACTIN are indicated. (D) External phenotypes of 2-month-old mice of the indicated genotypes. *Nr5a1-Cre*; *Sry-T-R26^{Kl/+}* XX mice ($n = 6$) developed as males but *Nr5a1-Cre*; *Sry-S-R26^{Kl/+}* XX mice ($n = 10$) did not. *Nr5a1-Cre*; *Sry-S-R26^{Kl/Kl}* XX mice ($n = 3$) developed as males. (E) Coimmunostaining profiles of SRY and NR5A1 in E11.5 gonads of the indicated genotypes. Subcellular localization of SRY-T and SRY-S are shown; for more detail, see fig. S6C. Data are representative of biological triplicate experiments. (F) Quantitation of SRY in NR5A1⁺ cells in E11.5 gonads of the indicated genotypes. Data are shown as means $\pm$ SD. Numbers of analyzed cells are shown in parentheses above each bar. *** $P < 0.001$ between the indicated groups, ANOVA with Tukey's post hoc test.



splicing results in a different 18 amino acids at the C terminus of SRY-S (Fig. 1D). We used CRISPR/Cas9 to establish the knock-in mouse line *Sry-T:3xFlag*, in which *Sry* exon2 is targeted to produce SRY-T with a triple Flag-epitope tag at the C terminus (fig. S4). We prepared protein lysates from embryonic day 11.5 (E11.5) *Sry-*

T:3xFlag gonads and enriched SRY using anti-bodies against the poly-Q sequence (10) (fig. S4A). Anti-FLAG immunoblotting identified a protein with the expected molecular weight of SRY-T-3xFLAG (~55 kDa) in the anti-SRY immune complexes (Fig. 1E). These results indicate that *Sry-T* is translated into a protein in vivo.

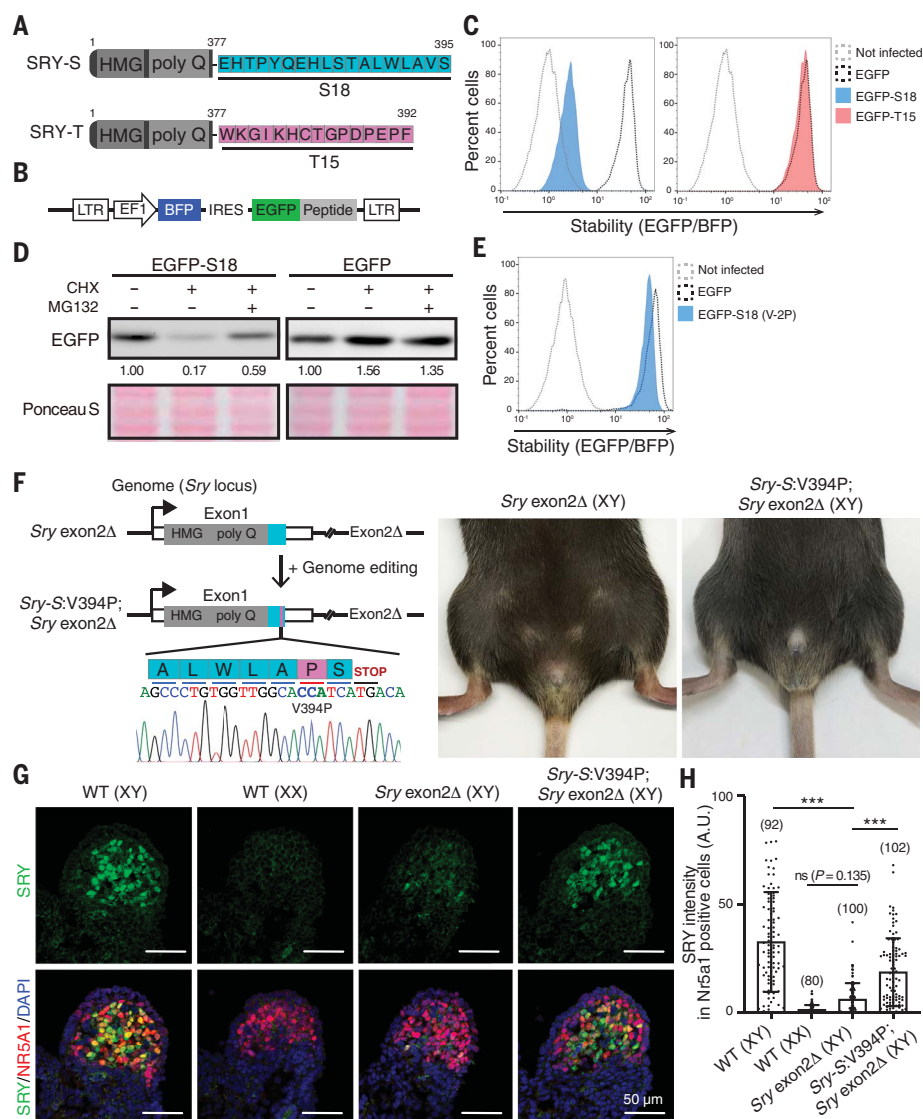
Next, we generated mice lacking the *Sry-T* transcript by deleting *Sry* exon2 (Fig. 2, A and B, and fig. S5, A and B). Elimination of *Sry-T* alone in XY mice resulted in complete male-to-female sex reversal (Fig. 2C and fig. S5, C to G), with a global gene expression profile of E11.5 XY *Sry-T*-deficient gonadal somatic cells

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Fig. 4. SRY-S carries a C-terminal degron.

(A) Specific C-terminal sequences of SRY-S and SRY-T were named S18 (blue) and T15 (red). (B) Lentivirus vector used for protein stability analysis. BFP and EGFP-fused polypeptides were translated from the same transcript. (C) Protein-destabilizing activity of S18 and T15 was gauged by the EGFP/BFP ratio using flow cytometry. (D) Western blot analysis of EGFP-S18 (left) and nonfusion EGFP (right) expressed in lentivirus-infected cells. Cells were treated with the protein synthesis inhibitor cycloheximide (CHX) for 12 hours in the presence or absence of the proteasome inhibitor MG132. Quantified protein levels are indicated. Equal protein loading was confirmed by Ponceau S staining. (E) V-2P substitution abolished S18 degron activity. (F) Editing of endogenous *Sry* to generate an *Sry*-S:V394P allele on an *Sry*-T-deficient background (left). *Sry*-S:V394P; *Sry* exon2Δ mice ($n = 4$ lines) developed as males (right). (G) Coimmunostaining profiles of SRY and NR5A1 in E11.5 gonads of the indicated genotypes. Data are representative of biological triplicate experiments. (H) Quantitation of SRY in NR5A1⁺ cells in E11.5 gonads of the indicated genotypes. Data are shown as means $\pm$ SD. Numbers of analyzed cells are shown in parentheses above each bar. *** $P < 0.001$, ns, not significant, between the indicated groups, ANOVA with Tukey's post hoc test.



similar to that of their XX counterparts (fig. S5H). *Sry-S* expression was maintained at normal levels in these mice (fig. S5I), suggesting that in wild-type (WT) XY mice, *Sry-T* and not the previously known *Sry-S* is necessary and sufficient for male sex determination.

To gain insights into this male-to-female sex reversal, we investigated total SRY protein expression in NR5A1⁺ gonadal somatic cells in WT and *Sry-T*-deficient XY embryos. SRY protein levels in *Sry-T*-deficient XY gonads were about one-sixth those of WT XY gonads (Fig. 2, D and E), suggesting that male-to-female sex reversal resulted from low SRY levels and that SRY-T is the predominant protein product of the *Sry* locus in XY gonads.

To compare the properties of SRY-T and SRY-S in detail, we established single-copy knock-in mouse lines conditionally expressing each isoform, which we named *Sry-T*R26^{KI/+} and *Sry-S*R26^{KI/+}, respectively. In this strategy (II), the coding DNA sequence (CDS) of either

Sry-T or *Sry-S* is expressed within the same context (i.e., copy number, integration site, promoter, and untranslated regions; Fig. 3A). The lines were then crossed with the *Nr5a1-Cre* transgenic line (6) to specifically express the *Sry* CDS in gonadal somatic cells. *Sry* mRNA levels in E11.5 XX knock-in gonads were comparable between the *Nr5a1-Cre; Sry-S-R26^{KI/+}* and *Nr5a1-Cre; Sry-T-R26^{KI/+}* mice (Fig. 3B). However, SRY protein levels in *Nr5a1-Cre; Sry-S-R26^{KI/+}* mice were approximately one-seventh those in *Nr5a1-Cre; Sry-T-R26^{KI/+}* mice (Fig. 3C), demonstrating that a posttranscriptional mechanism leads to the predominant expression of SRY-T over SRY-S.

Nr5a1-Cre; Sry-T-R26^{KI/+} XX mice developed as males, whereas *Nr5a1-Cre; Sry-S-R26^{KI/+}* XX mice did not (Fig. 3D and fig. S6, A and B). Immunofluorescence analysis indicated that SRY-T expression, not SRY-S expression, was sufficient to induce male development (Fig. 3, E and F). However, *Nr5a1-Cre; Sry-S-R26^{KI/KI}* XX mice,

which carry two expression cassettes of *Sry-S*, developed as males (Fig. 3D), indicating that doubling *Sry-S*-R26^{KI} expression surpassed the SRY protein-level threshold (Fig. 3, D to F) required to induce male development.

We speculated that the different C-termini of SRY-S (S18) and SRY-T (T15) (Fig. 4A) underlie the difference in postranscriptional regulation. We generated lentiviral constructs encoding blue fluorescent protein (BFP) and enhanced green fluorescent protein (EGFP) fused to either S18 or T15, both of which were translated from the same transcript (12) (Fig. 4B), and transduced them into human embryonic kidney 293T cells. The effect of S18 or T15 on protein stability was evaluated by examining the EGFP/BFP ratio. Whereas EGFP-T15 expression levels were similar to those of nonfusion EGFP (Fig. 4C, right), EGFP-S18 was present at <1/10 the nonfusion EGFP levels (Fig. 4C, left). In the presence of cycloheximide, EGFP-S18 was destabilized more rapidly than EGFP and EGFP-T15, which was

blocked by MG132 addition (Fig. 4D and fig. S7A). These data indicate that the C terminus of SRY-S contains a protein degradation motif, or degron.

Addition of a distal FLAG epitope counteracted the degradation of EGFP-S18 (fig. S7B), indicating that the C-terminal location of the S18 sequence is strictly required for its efficacy, consistent with a general property of C-terminal degrons (12). Further, a known subtype of C-terminal degrons carries valine at position -2 (V-2), and its substitution has been shown to abolish degron activity (12). Accordingly, we established a V-2P substitution in S18, which restored EGFP-S18 stability in transfected cells (Fig. 4E and fig. S7B). Thus, the S18 degron appears to be responsible for the differential protein expression of SRY-T and SRY-S.

To determine the significance of degron-mediated SRY-S instability in vivo, we edited the endogenous *Sry* gene using CRISPR/Cas9 to create mice expressing SRY-S with C-terminal mutations or substitutions on a *Sry-T*-deficient background and evaluated their sex development. XY *Sry-S:T15*; *Sry* exon2Δ mice, which express only SRY-S but with a degron-free T15 peptide instead of S18, developed as male (fig. S8). *Sry-S:V394P*; *Sry* exon2Δ mice, with a V-2P (amino acid position 394) substitution in SRY-S, also developed as male (Fig. 4F). Moreover, SRY protein levels were markedly elevated in *Sry-S:V394P*; *Sry* exon2Δ gonads compared with *Sry* exon2Δ gonads (Fig. 4, G and H), indicating that eliminating S18 degron activity stabilizes SRY-S to the level required for inducing male development. Finally, analysis of other mouse lines expressing SRY-S with an ordered series of C-terminal mutations on an *Sry-T*-deficient background confirmed that absence of the S18 degron is strictly required for SRY-S to activate the male pathway in vivo (fig. S8B and table S1). Thus, we conclude that the C-terminal degron prevents endogenously expressed SRY-S protein from accumulating to a level required for activating the male pathway, not only in *Sry-T*-deficient mice but also in WT XY mice.

Several previous studies showed that *Sry* transgenes encoding only SRY-S can induce testis development in XX mice under experimental conditions (1, 8, 9, 13). The sex reversal occurred in 25% (1) to 75% (8) of mice and was shown to correlate with transgene expression levels (9), which was in turn influenced by integration site and/or copy number (14). In support of this concept, heterozygous *Nr5a1-Cre*; *Sry-S-R26* knock-in XX mice did not develop as males but homozygotes did (Fig. 3D). Therefore, *Sry-S* transgene expression above a certain threshold can overcome the destabilizing effects of the degron. However, under physiological conditions, it is not SRY-S, but rather degron-free SRY-T, that is the bona fide testis-determining factor in the mouse.

Sry evolved from the single-exon gene *Sox3* (15, 16), and all previously characterized eutherian *Sry* genes are single exon. Therefore, our findings raise the important questions of how and why an additional exon arose in mice. We found that DNA sequences similar to the mouse *Sry-S* degron are conserved in *Sry* from other rodents (fig. S9A), but SRY was not destabilized in those species (fig. S9B) because the degron was blocked from translation due to preceding stop codons (fig. S9, A to D). In mice, a different mechanism has evolved to avoid degron translation through exonization of distal retrotransposon-derived sequences (Fig. 1C and fig. S3) and splicing to excise the degron (fig. S10). The apparent replacement of SRY-S by SRY-T to overcome degron activity provides an example of Y-chromosomal gene evolution by gain of function.

Our findings overturn the conclusion that has prevailed for 30 years regarding the structure of *Sry* in the mouse, the most intensively studied mammalian model of sex determination. Further study of SRY-T will establish whether exon2 confers new functionalities on SRY in addition to protein stability and will provide a more complete understanding of how the sex-determining cascade is activated in mice compared with other mammalian species.

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SUPPLEMENTARY MATERIALS

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Data S1 and S2
MDAR Reproducibility Checklist

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Two rather than one

For several decades, it has been believed that the mammalian sex-determining gene *Sry* contains a single exon. Miyawaki *et al.* have now identified a cryptic second exon of mouse *Sry*. Loss- and gain-of-function analyses revealed that the two-exon SRY (SRY-T), not the canonical single exon-encoded SRY (SRY-S), is the bona fide testis-determining factor. *Sry* exon2 is composed of retrotransposon-derived sequences. The SRY-S carboxyl terminus contains a degradation sequence (degron), whereas the SRY-T carboxyl terminus encoded in the *Sry* exon2 is degron free, thereby conferring protein stability on SRY-T.

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